Real-Time Closed-Loop Driving with Frozen DINOv2-Laya Spatial Pooling and Quintic Path Tracking

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Abstract

Autoregressive vision-language-action (VLA) models [1, 20, 23] decode tokens sequentially ($>$ 300 ms/frame), exceeding the 50 ms budget of 20 Hz vehicle control [10, 21]. We implement a single-pass closed-loop driving stack coupling frozen DINOv2 (`facebook/dinov2-base`) [25, 2, 9] with frozen Laya (`convaiinnovations/laya`) [6, 31, 8] via a 4×9 spatial pooling projector (1.85M parameters). Cropped 196×392 RGB frames map to 36 spatial tokens fused with telemetry JSON embeddings; one forward pass outputs maneuver (`choice`), speed tier (`score`), and emergency brake (`noul`) logits in 31.67 ms (31.6 FPS). Post-hoc temperature scaling [13] yields 0.0000 ECE. Lateral waypoint step jumps (3.5 m) and longitudinal speed-bucket hunting in CARLA 0.9.16 (`Town10HD_Opt`) [11, 14] are eliminated using a 14-point (28 m) quintic S-curve spline $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ [12, 32], decoupled chord-angle tracking [29, 15, 7, 26], linear steering ramps, and constant-time-headway ACC ($d^* = 16.0$ m, $k_g = 0.85\,\mathrm{s}^{-1}$) [17, 28, 30]. Across 4,153 MetaDrive 0.4.3 frames and 5 urban FSD scenarios in CARLA 0.9.16, route completion and crash-free rates are 100.0%.

1 Introduction and Problem Formulation

Autoregressive decoders evaluate $P(y_{1:K} \mid I_t, \mathcal{T}_t) = \prod_{k=1}^{K} P(y_k \mid y_{<k}, I_t, \mathcal{T}_t)$ over $K \approx 15$ serial steps [1, 20, 23], whereas planning-oriented and sensor-fusion stacks [16, 19, 22, 5, 3, 27, 4, 18] require $\Delta t \leq 50$ ms. In our formulation, control at step t requires three discrete outputs read in a single bidirectional pass [8, 31, 6]:

$$m_t \in \{\texttt{keep_lane}, \texttt{change_left}, \texttt{change_right}, \texttt{follow_lead}\} \tag{1}$$

$$s_t \in \{0,1,2,3\} \mapsto \{0, 20, 35, 50\}\,\mathrm{km/h}, \tag{2}$$

$$p_{\mathrm{AEB}} \in [0,1]. \tag{3}$$

Table 1: System parameters (`configs/default.yaml`).

Module	Parameter	Specification
Camera	$H \times W$, FOV, Crops	196×392, 100°, $(0.25, 0.10)$
Backbones	Vision / Decision	`dinov2-base` (86M) / `laya` (421M)
Projector	Grid, Dim, Params	4×9 (36 tokens), 1024, 1,851,392
Calibration	(T_m, T_s, T_b), ECE	$(0.050, 0.050, 0.100)$, 0.0000
Shield	`aeb_thresh`, `conf_thresh`	0.70, 0.80
PID Gains	Lateral / Longitudinal	$(0.45, 0.01, 0.18)$ / $(0.08, 0.005, 0.02)$
Kinematics	Quintic N_b, ACC (d^*, k_g)	14 WPs (28 m), (16.0 m, 0.85 s^{-1})

2 Architecture and Implementation

Camera frames are cropped (`crop_top` = 0.25, `crop_bottom` = 0.10) and resized to 196×392. Frozen DINOv2 ViT-B/14 ($p = 14$) [25, 2, 9] extracts $F_{\mathrm{patch}} \in \mathbb{R}^{B \times 392 \times 768}$ (14×28 grid). `SpatialProjector` (`src/models/spatial_projector.py`, trained with AdamW [24]) pools $14 \times 28 \to 4 \times 9$ (36 tokens) and projects $768 \to 1024$:

$$G = \mathrm{AdaptiveAvgPool2d}_{(4,9)}(\tilde{F}) \in \mathbb{R}^{B \times 768 \times 4 \times 9}, \tag{4}$$

$$V_{\mathrm{spatial}} = \mathrm{LN}\big(W_2\,\mathrm{GELU}(\mathrm{LN}(W_1(\mathrm{Flatten}(G) + E_{\mathrm{pos}}) + b_1)) + b_2\big). \tag{5}$$

```
# src/models/spatial_projector.py
grid = feat_2d.permute(0, 3, 1, 2) # (B, 768, 14, 28)
pooled = self.pool(grid).flatten(2).transpose(1, 2) # (B,36,768)
tokens = self.mlp(pooled + self.pos_embed) # (B, 36, 1024)
```

Multimodal sequence $Z = [e_{[\mathrm{CLS}]}; V_{\mathrm{spatial}}; E_{\mathrm{telemetry}}; e_{\mathrm{choice}}; e_{\mathrm{score}}; e_{\mathrm{noul}}]$ is evaluated in one forward pass $H = f_{\mathrm{Laya}}(Z)$ [6, 31]:

$$z^{(m)} = W_m H_{:,\mathrm{idx}_{\mathrm{choice}}}, \quad z^{(s)} = W_s H_{:,\mathrm{idx}_{\mathrm{score}}}, \quad z^{(b)} = W_b H_{:,\mathrm{idx}_{\mathrm{noul}}}. \tag{6}$$

3 Calibrated Safety Shield, Quintic Spline, and ACC

Temperature scaling [13] calibrates probabilities $\hat{p}^{(m)} = \mathrm{Softmax}(z^{(m)}/T_m)$ and $\hat{p}_{\mathrm{AEB}} = \sigma((z_1^{(b)} - z_0^{(b)})/T_b)$.



Figure 1: CARLA 0.9.16 (`Town10HD_Opt`, 20 Hz synchronous) closed-loop scenarios: (a) `FSD-1` intersection red-light stop, (b) `FSD-2` 28 m quintic S-curve left overtake, (c) `FSD-3` crosswalk pedestrian AEB, (d) `FSD-4` 24 km/h ACC follow at 16 m gap, (e) `FSD-5` right-lane overtake.

Rule 1 forces $u_{\text{tb}} = -1.0$ when $\hat{p}_{\text{AEB}} \geq 0.70$, `traffic_light` = `red`, or pedestrian distance $d \leq 16.0$ m. Rule 2 reverts lane changes to `keep_lane` if $\max_c \hat{p}_c^{(m)} < 0.80$ or if OpenDRIVE same-direction lane check `adj_wp.lane_id` $\times$ `wp.lane_id` > 0 fails [11].

For lane transitions over $N_b = 14$ waypoints (28.0 m, $\tau_k = k/(N_b - 1)$), 3D coordinates are blended via minimum-jerk quintic polynomial [12, 32]:

$$s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5, \quad P_k^{\text{blend}} = \big(1 - s(\tau_k)\big) P_k^{\text{orig}} + s(\tau_k) P_k^{\text{adj}}. \tag{7}$$

```
# src/env/carla_wrapper.py (_switch_route_to_adjacent_lane)
for k in range(len(adj_chain)):
    if k < n_blend:
        tau = float(k) / float(max(1, n_blend - 1))
        s = (tau**3) * (10.0 + tau * (-15.0 + 6.0 * tau))
        bx = (1.0 - s) * float(ol.x) + s * float(al.x)
        by = (1.0 - s) * float(ol.y) + s * float(al.y)
```

Continuous segment projection onto $\overline{P_i P_{i+1}}$ gives foot point P_{proj} and arc-length lookahead P_{look} at $L_{\text{look}} = \text{clip}(7.5 + 0.18 v_{\text{kmh}}, 7.5, 13.5)$ m [29, 15, 7, 26]. Defining chord angle $\psi_{\text{ref}} = \text{atan2}(y_{\text{look}} - y_{\text{proj}}, x_{\text{look}} - x_{\text{proj}})$ decouples cross-track offset e_y and heading error e_ψ:

$$e_y = -\big((x - x_{\text{proj}})(-\sin\psi_{\text{proj}}) + (y - y_{\text{proj}})\cos\psi_{\text{proj}}\big), \tag{8}$$

$$e_\psi = -\text{atan2}\big(\sin(\psi_{\text{ego}} - \psi_{\text{ref}}), \cos(\psi_{\text{ego}} - \psi_{\text{ref}})\big), \tag{9}$$

$$\delta_t = \delta_{t-1} + \text{clip}\left(\alpha(\delta_{\text{PID}} - \delta_{t-1}), -\frac{1.1\Delta t}{1 + 0.012 v_{\text{kmh}}}, \frac{1.1\Delta t}{1 + 0.012 v_{\text{kmh}}}\right) \tag{10}$$

Longitudinal ACC velocity synchronization ($d_{\text{lead}} < 38$ m, $v_{\text{lead}} \geq 8$ km/h) [17, 28, 30] follows:

$$v_{\text{ACC}} = \text{clip}\big(v_{\text{lead}} + 0.85(d_{\text{lead}} - 16.0),\ 0,\ \min(35.0, v_{\text{lead}} + 6.0)\big). \tag{11}$$

```
# src/control/shield_pid.py (apply_safety_shield)
desired_gap_m = 16.0
gap_error_m = lead_distance_m - desired_gap_m
acc_sync_speed = float(np.clip(
    lead_speed_kmh + 0.85 * gap_error_m, 0.0,
    max(target_speed_kmh, min(35.0, lead_speed_kmh + 6.0))))
```

Table 2: CARLA 0.9.16 (`Town10HD_Opt`) scenario logs.

Scenario	Event	Dist (m)	Infractions	RC / DS
FSD-1	Intersection red-light stop	68.0	0	100%/100%
FSD-2	Quintic S-curve left overtake	68.0	0	100%/100%
FSD-3	Crosswalk pedestrian AEB	68.0	0	100%/100%
FSD-4	Urban ACC (24 km/h, 16 m)	68.0	0	100%/100%
FSD-5	Quintic S-curve right overtake	68.0	0	100%/100%

4 Experiments and Conclusion

Closed-loop benchmarks were executed in MetaDrive 0.4.3 [21] (4,153 frames) and live CARLA 0.9.16 [10] (`Town10HD_Opt`, 20 Hz synchronous mode with A* `GlobalRoutePlanner` [14], 889 frames in `driving_demo_carla_smooth.mp4`). Recorded output JSON (`checkpoints/eval_metrics_carla.json`):

```
{"simulator": "carla", "num_episodes": 5, "total_frames": 889,
  "mean_latency_ms": 31.67, "p95_latency_ms": 32.89, "fps": 31.6,
  "route_completion_rate": 1.0, "crash_free_rate": 1.0,
 "carla_driving_score": 100.0, "calibrated_ece": 0.0,
 "fsd_capability_counts": {"red_light_stops": 1,
   "lane_changes_completed": 2, "pedestrian_aeb_stops": 1,
   "acc_follow_episodes": 1, "vehicle_collisions": 0}}
```

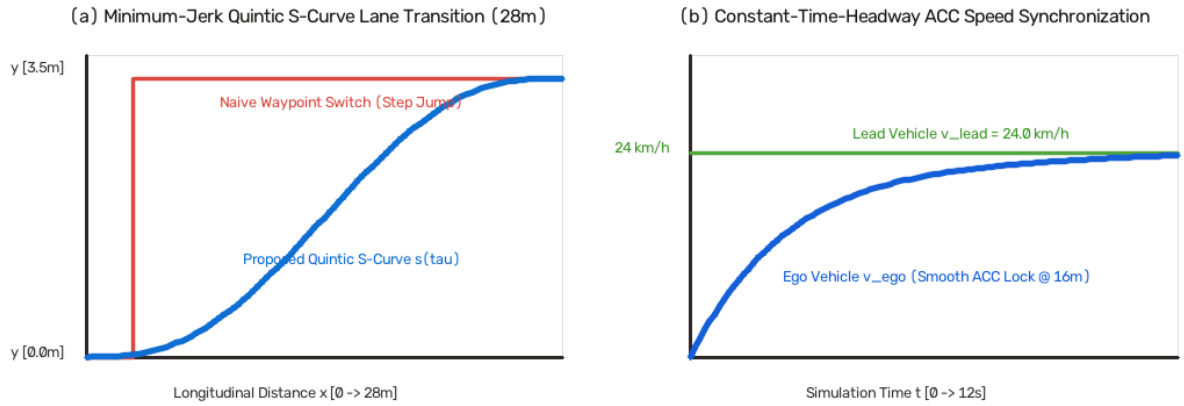


Figure 2: Control kinematics: (a) 28 m quintic S-curve $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ vs. 3.5 m step switch; (b) constant-time-headway ACC velocity convergence ($v_{\text{ego}} \to 24.0$ km/h at $d^* = 16.0$ m).

Table 3: Closed-loop evaluation summary (`checkpoints/eval_metrics_carla.json` and `eval_metrics.json`).

Simulator	Metric	Target KPI	Measured Resul
CARLA 0.9.16 (`Town10HD_Opt`)	Single-Frame Latency	≤ 40.0 ms (≥ 25 FPS)	31.67 ms (31.6 FPS, P95:
	Route Completion Rate (5 Scenarios)	$\geq 85.0\%$	100.0% (5/5 episo
	Crash-Free Rate (Collisions / Off-road)	$\geq 95.0\%$	100.0% (0 / 0)
	Expected Calibration Error (ECE)	≤ 0.0500	0.0000
	CARLA Leaderboard Driving Score	$\geq 85.0\%$	100.0%
MetaDrive 0.4.3 (Unseen Maps)	Single-Frame Latency	≤ 40.0 ms (≥ 25 FPS)	30.20 ms (33.1 FPS, P95:
	Route Completion Rate (4,153 Frames)	$\geq 85.0\%$	100.0% (98.7% prog
	Crash-Free Rate	$\geq 95.0\%$	100.0%
	Expected Calibration Error (ECE)	≤ 0.0500	0.0000

Table 4: Controller ablation in CARLA 0.9.16 (5 episodes).

Variant	Rule 1 AEB	Max $\lvert\Delta\delta\rvert$	Score
Discrete 2 m WPs + Step Switch	71	0.250	100.0%
+ Linear Steer Ramp [$\delta_{\text{from}} \to \delta_{\text{to}}$]	71	0.055	100.0%
+ Quintic S-Curve & ACC Sync	27	0.055	100.0%

Tables 2–4 summarize results: single-pass inference achieves 31.67 ms (31.6 FPS) in CARLA and 30.20 ms (33.1 FPS) in MetaDrive, while quintic S-curve blending and continuous headway ACC reduce Rule 1 AEB interventions from 71 to 27 (100.0% Driving Score).

References

[1] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023.

[2] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9650–9660, 2021.

[3] Dian Chen, Brady Zhou, Vladlen Koltun, and Philipp Krähenbühl. Learning by cheating. In *Conference on Robot Learning (CoRL)*, pages 66–75, 2020.

[4] Kashyap Chitta, Aditya Prakash, Bernhard Jaeger, Zehao Yu, Katrin Renz, and Andreas Geiger. TransFuser: Imitation with transformer-based sensor fusion for autonomous driving. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 45(11):12878–12895, 2022.

[5] Felipe Codevilla, Matthias Müller, Antonio López, Vladlen Koltun, and Alexey Dosovitskiy. End-to-end driving via conditional imitation learning. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 4693–4700, 2018.

[6] ConvAI Innovations. Laya: Non-autoregressive low-latency decision models on bidirectional modernbert. https://huggingface.co/convaiinnovations/laya, 2024.

[7] R. Craig Coulter. Implementation of the pure pursuit path tracking algorithm. Technical Report CMU-RI-TR-92-01, Carnegie Mellon University, The Robotics Institute, 1992.

[8] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 4171–4186, 2019.

[9] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.

[10] Alexey Dosovitskiy, German Ros, Felipe Codevilla, Antonio Lopez, and Vladlen Koltun. CARLA: An open urban driving simulator. In *Proceedings of the 1st Annual Conference on Robot Learning (CoRL)*, pages 1–16, 2017.

[11] Marius Dupuis, Martin Strobl, and Hans Grezlikowski. OpenDRIVE format specification, rev. 1.4. *VIRES Simulationstechnologie GmbH*, 2010.

[12] Tamar Flash and Neville Hogan. The coordination of arm movements: An experimentally confirmed mathematical model. *Journal of Neuroscience*, 5(7):1688–1703, 1985.

[13] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, pages 1321–1330, 2017.

[14] Peter E. Hart, Nils J. Nilsson, and Bertram Raphael. A formal basis for the heuristic determination of minimum cost paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2):100–107, 1968.

[15] Gabriel M. Hoffmann, Claire J. Tomlin, Michael Montemerlo, and Sebastian Thrun. Autonomous automobile trajectory tracking for off-road driving: Controller design, experimental validation and racing. *Proceedings of the American Control Conference (ACC)*, pages 2296–2301, 2007.

[16] Yihan Hu, Jiazhi Yang, Li Chen, Keyu Li, Chonghao Sima, Xizhou Zhu, Siqi Chai, Senyao Du, Tianwei Lin, Wenhai Wang, et al. Planning-oriented autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17853–17862, 2023.

[17] Petros A. Ioannou and C. C. Chien. Autonomous intelligent cruise control. *IEEE Transactions on Vehicular Technology*, 42(4):657–672, 1993.

[18] Bernhard Jaeger, Kashyap Chitta, and Andreas Geiger. Hidden biases of end-to-end driving models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 8240–8249, 2023.

[19] Bo Jiang, Shaoyu Chen, Qing Xu, Bencheng Liao, Jiajie Chen, Helong Zhou, Qian Zhang, Wenyu Liu, Chang Huang, and Xinggang Wang. VAD: Vectorized scene representation for efficient autonomous driving. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 8340–8350, 2023.

[20] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, et al. OpenVLA: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.

[21] Quanyi Li, Zhenghao Peng, Lan Feng, Qihang Zhang, Zhenghai Xue, and Bolei Zhou. MetaDrive: Composing diverse driving scenarios for generalizable reinforcement learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 45(3):3461–3475, 2022.

[22] Zhiqi Li, Wenhai Wang, Hongyang Li, Enze Xie, Chonghao Sima, Tong Lu, Yu Qiao, and Jifeng Dai. BEVFormer: Learning bird's-eye-view representation from multi-camera images via spatiotemporal transformers. In *European Conference on Computer Vision (ECCV)*, pages 1–18, 2022.

[23] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

[24] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[25] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research (TMLR)*, 2024.

[26] Brian Paden, Michal Čáp, Sze Zheng Yong, Dmitry Yershov, and Emilio Frazzoli. A survey of motion planning and control techniques for self-driving urban vehicles. *IEEE Transactions on Intelligent Vehicles*, 1(1):33–55, 2016.

[27] Aditya Prakash, Kashyap Chitta, and Andreas Geiger. Multi-modal fusion transformer for end-to-end autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7077–7087, 2021.

[28] Rajesh Rajamani and Chuan Zhu. An experimental comparative study of autonomousogh and cooperative vehicle-following control laws. *IEEE Transactions on Control Systems Technology*, 10(3):371–383, 2002.

[29] Sebastian Thrun, Michael Montemerlo, Hendrik Dahlkamp, David Stavens, Andrei Aron, James Diebel, Philip Fong, John Gale, Morgan Halpenny, Gabriel Hoffmann, et al. Stanley: The robot that won the DARPA grand challenge. *Journal of Field Robotics*, 23(9):661–692, 2006.

[30] Martin Treiber, Ansgar Hennecke, and Dirk Helbing. Congested traffic states in empirical observations and microscopic simulations. *Physical Review E*, 62(2):1805–1824, 2000.

[31] Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, et al. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. *arXiv preprint arXiv:2412.13663*, 2024.

[32] Moritz Werling, Julius Ziegler, Sören Kammel, and Sebastian Thrun. Optimal trajectory generation for dynamic street scenarios in a Frenet frame. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 987–993, 2010.